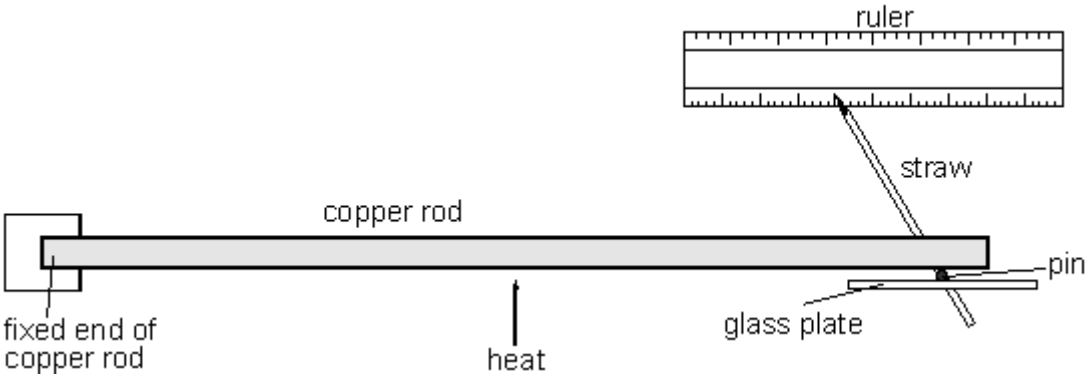
Q1.

Sarah wanted to find out how the length of a copper rod changes when it is heated.
She fixed one end of the rod and allowed the other end to roll on a pin which had been put on a glass plate.
A drinking straw, attached to the pin , showed the movement of the end of the rod.



(a) (i) What happens to the motion of the copper atoms in the rod as it is heated?

.....

.....

1 mark

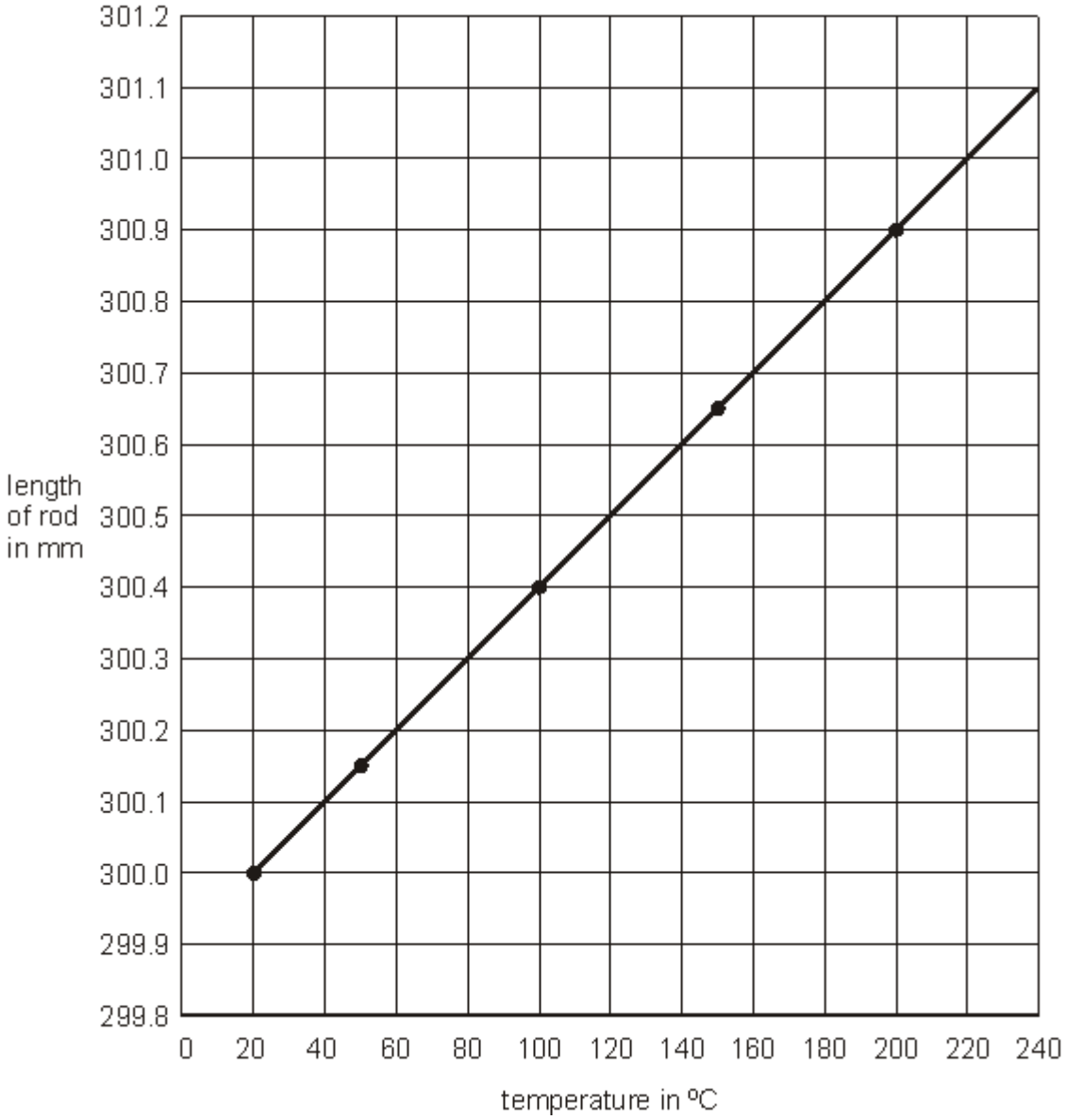
(ii) What happens to the average distance between the atoms of copper when the rod is heated?

.....

.....

1 mark

(b) The graph shows the length of a copper rod at different temperatures.
The rod was 300.0 mm long at room temperature.



(i) At what temperature has the rod increased its length by 1.0 mm?

.....°C
1 mark

(ii) The rod was 300.0 mm long at room temperature. What will its length become if it is placed in melting ice at 0°C?

..... mm
1 mark
Maximum 4 marks

Q1.

(a) (i) any **one** from

- they vibrate further
  *accept 'they vibrate more **or** faster'*
- they move faster
  accept 'they go faster'
  do not accept 'they move about more'
  ***or** 'they collide more'*

1 (L6)

(ii) it increases
*accept 'it gets bigger' **or** 'they move further apart'*

1 (L6)

(b) (i) 220

1 (L5)

(ii) 299.9

1 (L5)

[4]